

Progress Report

Project: Data-Driven Publishing Strategy for Steam Game Releases Client: Pixel Publishing (Director of Publishing)

Stage: Milestone 1, Exploratory Data Analysis Date: 5 August 2026

So far I have validated the dataset, scoped the sample, defined and justified the target variable, and looked at the first two predictors named in the client's first question (genre and price).

The dataset came in two format, games.csv (125855 games) and games.json (137301 games). I picked the CSV because it matches the 125855 figure the client proposal cites, and because the fields only in the JSON do not help much. The tag vote counts are assigned by players after release so they would leak the outcome, and the extra description fields repeat text the CSV already has.

The CSV header is broken. It has 39 column names but every data row has 40 values, because the source file wrote Discount and DLC count without a comma between them. Loaded normally, pandas quietly shifts every column by one and gives no error. Before I fixed it, the col labelled Release date was holding ownership ranges.

I fixed it by relabelling the 39 col with their real names which also pushed AppID into the index where it cannot be mistaken for a feature.

The same fault explains something odd I saw when opening the file in a spreadsheet, where the col labelled Movies looks full of image links. Those links are screenshot URLs sitting under a label that slid one position. The Movies field is actually empty in all 125855 rows.

I built a summary table with data type, null count, fill rate, unique count, most common value, and what share of rows hold that value. The last two col turned out to be necessary, because several fields say 100% complete while carrying no information. User score is the value 0 in 99.97% of rows, Required age in 98.98%, and Windows is True in 99.97%. Fill rate on its own would have marked all three as usable.

Estimated owners sorts games into 14 ordered ranges. The distribution is very lopsided. About 79.68% of games sit at or below 20000 owners, and only 220 games (~0.175%) go above 5000000.

I defined success as 50000 owners or more. This keep bar that actually mean something for a publisher releasing 3 to 6 titles a year while still leaving enough positive cases to model. It give 14174 successful games, or 12.02% of my working sample.

That number matter later. A model that predicts not successful for every title would score about 88% accuracy while being useless, so I cannot judge my model on accuracy.

Five metadata col Genres, Categories, Developers, Publishers, About the game were each missing a similar number of rows. When I checked, I found 7983 rows missing all five at once. Of those, 7858 (98.4%) have Playtest in the title, and all 7983 sit in the 0-0 ownership group with no exceptions. I removed them which leaves a working sample of 117872 games.

The reason I removed them matter actually cause I removed them for missing metadata not for low ownership. The client proposal says post launch outcomes must not decide eligibility, because filtering on the outcome biases the analysis. So I kept the other 16596 games that show zero owners but do have full metadata. Those are real commercial failures, and dropping them would leave me training only on games that got some traction, which would give the client advice that is too optimistic.

What I have found so far

Genre. The 2922 distinct values in Genres are combinations. There are only 33 real genres, and games carry 2.88 on average. Success rates run from Massively Multiplayer at 23.46% down to Casual at 7.73%, so about a 3 times spread. But 7 of the 13 large genres fall between 10% and 14%, crowded around the 12.02% base rate, so most of that spread comes from a few genres at the extremes.

Price. Free games show up at both ends of the ownership scale and are rare in the middle. They are 52.6% of the zero owner games, drop to 8.0% in the 0 to 20000 group, then climb steadily to about 18% in the higher tiers. So free release looks like a high risk high reward choice. Median price by genre does not explain the genre ranking, since the most successful genre is free and the most expensive genre is close to the least successful.